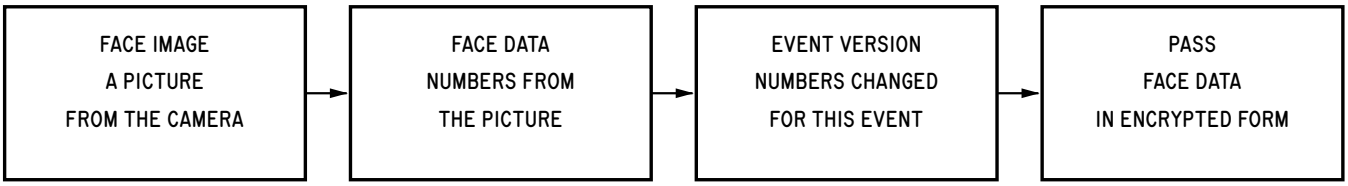
SECTION 1
FACE DATA AND MATCH ERRORS

FACE WORDS / MATCH ERRORS: PAGE 1 · PASS SECURITY / OFFLINE / TEST CLAIMS: PAGE 2

1.1 WHAT IS FACE DATA?

A face image is a picture. Face data is a set of numbers that a computer makes from that picture. A face template is another name for this set. Private Pass changes the numbers for each event before it puts them in a pass (figure 1).



PPS-501A

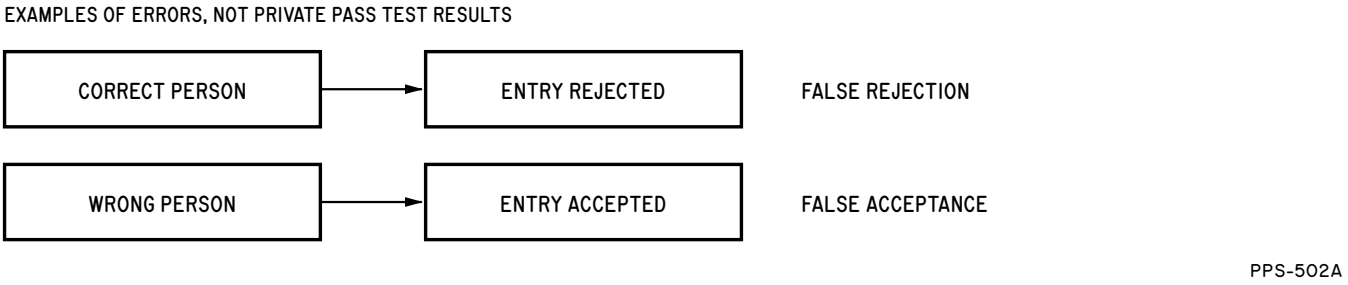
Figure 1. A Picture Becomes Data in a Pass

1.2 WHAT DOES 1:1 MEAN?

A 1:1 check compares one person with one pass. Example: Alex shows a pass. The gate compares Alex's face with the face data in that pass. It does not search a list of all visitors.

1.3 WHAT DOES THE MATCH LIMIT DO?

The match limit sets how close the two sets of numbers must be. With a small permitted difference, the gate can reject the correct person. With a large permitted difference, it can accept the wrong person. These are different errors (figure 2).



PPS-502A

Figure 2. Two Possible Errors in a Face Check

Table 1. Face Terms: Quick Lookup

Term in the talk	Meaning
Pass	A digital ticket with face data.
Gate	The device for the entry check.
Enrollment	Connects a person's face to a pass.
Template	Numbers made from a face image.
Embedding	Another name for a template.
Spoof attack	An attempt to enter with a fake face.

NOTE

Face data can show private information about a person. Numbers do not remove every privacy risk.

FIND AN EXPLANATION

1 Find the term in a heading.
2 Read the short explanation.
3 Use the example or drawing.
4 Use the table for other terms.